Practical 10 - Code

Lex file: pract10.l

*pract10.l

```
Name - Mohammed Zaid Shaikh
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Batch A2

%{
#include "y.tab.h"
%}

%%
"for"    { return FOR; }
[a-zA-Z]+  { return ID; }
[0-9]+    { return NUM; }
"=="|"="|"+"|"|-"|"|**"|"=" { return RELOP; }
"++"   { return INC; }
"--"    { return DEC; }
[ \t\n]+   ;
.     { return yytext[0]; }
%%

int yywrap()
{
    return 1;
}
```

Yacc/Bison file: pract10.y

*pract10.y

```
Name: Mohammed Zaid Shaikh
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Batch A2

%{
#include <stdio.h>

int yylex(void);
int yyerror(const char *s);
%}

%token FOR ID NUM RELOP INC DEC

%%
input : stmt
      ;

stmt  : FOR '(' ID '=' NUM ';' ID RELOP NUM ';' ID INC ')'
      | FOR '(' ID '=' NUM ';' ID RELOP NUM ';' ID DEC ')'
      | ID RELOP ID '?' ID ':' ID
      | ID RELOP NUM '?' ID ':' ID
      | ID RELOP ID '?' NUM ':' NUM
      | ID RELOP NUM '?' NUM ':' NUM
      ;
%%

int yyerror(const char *s)
{
    printf("Invalid statement\n");
    return 1;
}

int main()
{
    printf("Enter a for loop or ternary expression:\n");

    if(yyparse() == 0)
        printf("Valid statement\n");

    return 0;
}
```

Practical 10 - Output

```
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter a for loop or ternary expression:
for(i=0;i<25;i++)
Valid statement
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter a for loop or ternary expression:
a>9?a:g
Valid statement
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter a for loop or ternary expression:
for(i=0,i<n,i++)
Invalid statement
zaid787@LAPTOP-ID0Q783M:~$ ./a.out
Enter a for loop or ternary expression:
a>b!a:b
Invalid statement
zaid787@LAPTOP-ID0Q783M:~$
```